UNIT – IV

Industrial Structures

Syllabus: Design of Roof Trusses – Loads on Trusses – Purlin Design Using Angle and Channel Sections – Truss Design, Design of Joints and End Bearings – Design of Gantry Girder – Introduction to Pre-Engineered Buildings.

What's inside these notes

- Industrial buildings & roof truss types
- Loads on trusses (DL, LL, wind IS 875) & combinations
- Purlin design – angle & channel (worked example)
- Truss analysis & member design, joints, gusset plates
- End bearings (fixed & roller)
- Gantry girder: loads, section, checks, example
- Pre-Engineered Buildings (PEB)

Exam tip: Truss answers need a neat sketch with panel points, loads on joints, a table of member forces (+ tension / – compression) and then member + joint design.

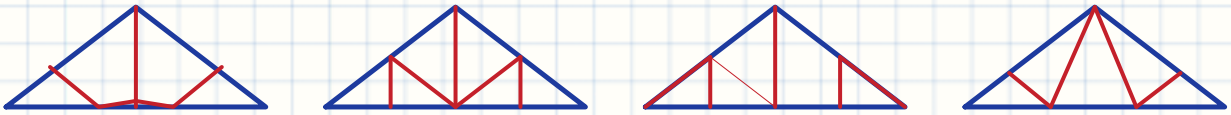
Codes used: IS 800:2007 (Limit State Method), IS 808 / SP-6(1) steel tables, IS 875 (loads), IS 2062 (steel grades). Take $E = 2\times10^5$ MPa, $f_y = 250$ MPa, $f_u = 410$ MPa unless stated.

1. Industrial Structures – Introduction

Single-storey **industrial buildings** – factories, workshops, warehouses, power houses, hangars. Main parts:

- **Roof covering** (GI / AC / colour-coated sheets) → **purlins** → **roof trusses** → columns / stanchions → foundations
- **Bracings** (vertical, horizontal – wind, stability), **girts / sheeting rails** for walls, **gantry girders** with crane
- **Bay** = spacing of trusses (4–6 m economical) ; span 10 – 40 m.

Roof truss types



Common roof truss shapes

- **Fink / Fan** – small to medium spans ; short web members in compression
- **Howe** – diagonals in compression, verticals tension ; **Pratt** – diagonals tension (long members in tension → economic)
- **North-light**, **Mansard**, **Warren**, **Saw-tooth**, **Bowstring / parallel-chord** (flat roofs)
- Rise : **1/5 – 1/4 span** typical ; **slope ≥ 15°** for AC sheet roofs (or 1 in 3 for tiles) ; Pratt/Howe for 6–15 m span.
- Panel length chosen so purlins fall at **panel points** (no bending in rafters) ~ 1.5–2.5 m.

Sheeting

Sheet	Weight (approx.)	Remarks
GI corrugated	0.13 – 0.15 kN/m²	light, quick
AC sheet	0.17 kN/m²	brittle, min. slope 15°
Colour-coated profiled	0.05–0.10 kN/m²	PEB, modern

2. Loads on Roof Trusses

(a) Dead loads (DL)

Weight of sheeting + fixtures (0.15–0.25 kN/m²), purlins (≈ 0.10 kN/m²), **truss self-weight ≈ (L/3 + 5) × 10 N/m² of plan area** (L = span in m ≈ 0.12–0.20 kN/m²), bracings.

(b) Imposed (live) load – IS 875 (Part 2)

- Roof **not accessible** except for maintenance : **0.75 kN/m²** (slope ≤ 10°) ; reduce by **0.02 kN/m² per degree** above 10° (min. **0.4 kN/m²**), measured on **plan area**.

- Plus 0.9 kN concentrated at the mid-span of purlins (check) ; dust/snow loads where relevant.

(c) Wind load – IS 875 (Part 3)

$$V_z = V_b\, k_1\, k_2\, k_3\, k_4 \text{ (m/s)}$$

$$p_z = 0.6\, V_z^2 \text{ (N/m}^2\text{)} ; \; p_d = K_d\, K_a\, K_c\, p_z$$

$$F = (C_{pe} - C_{pi})\, A\, p_d \text{ (normal to surface)}$$

- V_b = basic wind speed (zone) ; k_1 risk coeff ; k_2 terrain & height ; k_3 topography ; k_4 importance (cyclonic).
- C_{pe} external pressure coefficients (slope, h/w ratio, windward/leeward) ; C_{pi} internal : **±0.2** (low permeability), **±0.5** (medium), **±0.7** (large openings).
- Critical case for light roofs: **suction (uplift)** may exceed DL → **net uplift** governs purlin & fastener design & stability (0.9 DL + 1.5 WL).

(d) Other: seismic (light roofs - small), snow, crane (columns), temperature.

Load combinations (strength)

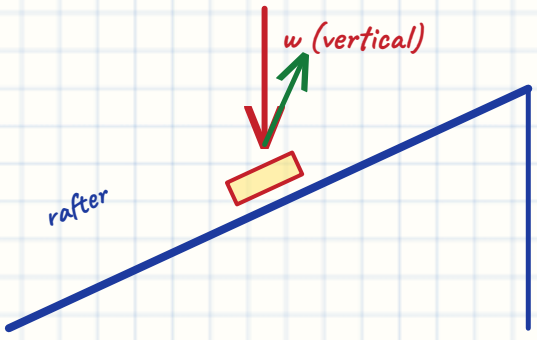
$$1.5\,(DL + LL) ; \; 1.2\,(DL + LL + WL) ; \; 1.5\,(DL + WL) ; \; 0.9\,DL + 1.5\,WL$$

Wind is applied **normal to the sloping roof surface**; resolved into joint loads : P = p × (purlin spacing along slope) × (truss spacing).

> Panel-point loads : W_{joint} = w (kN/m²) × (c/c truss spacing) × (c/c purlin spacing on plan). Apply **half loads** at eave/ridge joints.

3. Design of Purlins

Purlins = beams spanning between trusses, carrying sheeting load. Sections : **angle (ISA)** (small spans ≤ 4 m), **channel (ISMC)**, **I-section / tubular / cold-formed Z / C** (larger spans).



θ = roof slope

$w_n = w \cos\theta$ (⊥ roof) → M_x

$w_p = w \sin\theta$ (∥ roof) → M_y

sag rods at L/3 (or L/2) reduce M_y

both axes: $M_x/M_{dx} + M_y/M_{dy} \le 1$

span = truss spacing

Purlin on an inclined rafter – load components

Load on purlin (per metre length)

- Spacing along slope s ; $w = (DL_{sheet} + DL_{purlin}) + LL$ (plan → × cosθ)… ; factored $w_u = 1.5\,w$ (or wind combination, **w_n only**).
- Component **normal to roof** : $w_n = w\cos\theta$ → bending about **major axis** (M_x)
- Component **parallel to roof** : $w_p = w\sin\theta$ → bending about minor axis (M_y) (reduced by **sag rods**).

$$M_x = w_n L^2/10 \text{ (continuous / approx.) or } w_n L^2/8 \text{ (SS)} ; M_y = w_p (L/n)^2/10 \ (n = \text{no. of sag-rod spaces})$$

- **Wind load** acts ⊥ roof (no w_p) — check **uplift** with DL (0.9DL + 1.5WL).

Design steps

- 1. Compute w_n, w_p (factored). 2. M_x, M_y, shear.
- 3. Trial: $Z_{p,req} = M_x \gamma_{m0}/f_y$ (×1.2–1.4 for biaxial) → choose ISMC / ISA (also with deflection: **$\delta \le L/150$ (elastic cladding)**).
- 4. **Biaxial check**: $M_x/M_{dx} + M_y/M_{dy} \le 1.0$ (or $M_y/(Z_{py} f_y/\gamma_{m0})$ …) ; $M_{dy} \le 1.2\, Z_{ey} f_y/\gamma_{m0}$.
- 5. Shear (low), deflection (separately for both axes: $\delta = \sqrt{(\delta_x^2 + \delta_y^2)} \le L/150$).
- 6. Check **web crippling** at the bearing and **sag rod** (tension = $w_p L \times k$; $\varphi \ge 12$ mm).
- 7. Check **LTB** – roof sheeting prevents it if fixed to top flange (assume supported).

Why sag rods? w_p acts along slope; channel/angle is weak about minor axis → rods at L/3 (or mid) pick up w_p and pass it to the **ridge purlin / tie** . Sag rod diameter ≥ 12 mm.

Example 1 – Purlin loads and design moments

Truss spacing $L = 4$ m ; purlin spacing along slope $s = 1.5$ m ; roof slope $\theta = 20°$; AC sheets.

DL : sheet + fixtures $0.25 \text{ kN/m}^2 \times 1.5 = 0.375$ kN/m ; purlin self-wt 0.07 kN/m → **0.445 kN/m** (on slope; per m run)

LL (plan) : $0.75 - 0.02\,(20 - 10) =$ **0.55 kN/m²** → on purlin = $0.55 \times 1.5 \times \cos 20° = 0.775$ kN/m (vertical)

Total vertical $w = 0.445 + 0.775 =$ **1.22 kN/m** ; factored $w_u = 1.5 \times 1.22 =$ **1.83 kN/m**

$w_n = 1.83\cos 20° =$ **1.72 kN/m** ; $w_p = 1.83\sin 20° =$ **0.63 kN/m**

$M_x = 1.72 \times 4^2/10 =$ **2.75 kN·m** (continuous) ; with 1 sag rod at mid-span : $M_y = 0.63 \times 2^2/10 =$ **0.25 kN·m**

Required $Z_{p,x} \approx 2.75\times10^6 \times 1.1/250 = 12.1$ cm³ ; with biaxial factor ×1.4 ≈ 17 cm³ → very light – deflection governs : $\delta = 5 w_n L^4/(384EI) \le L/150 = 26.7$ mm → $I \ge 5 \times (1.72/1.5 = 1.15) \times 4000^4/(384 \times 2\times10^5 \times 26.7) = 7.2\times10^5$ mm⁴ = 72 cm⁴ → **ISMC 75** ($I_{xx} \approx 76$ cm⁴) or ISA 90×90×6 → check biaxial & deflection with exact table values.

4. Analysis and Design of Truss Members

Analysis

- Assume **pin-jointed truss**, loads only at joints. Reactions by statics (check uplift for wind).
- **Method of joints** or **method of sections** or graphical (Maxwell–Bow); tabulate member forces for **DL, LL, WL (left/right)** ; take critical combination ; (+ tension, − compression). Reversal in members possible under wind → design for both.
- Secondary bending ignored if members are concentric at joints.

Members & minimum sizes

Member	Section	Notes
Top chord (rafter)	**double angle** (back to back) or tee	compression ; purlins at panel points ; r_{min} governs
Bottom chord	double angle	tension; net area with holes
Web struts	single/double angle	long struts: compression
Web ties	single angle / rods	tension

Min. angle size **ISA 50 × 50 × 6** (thickness ≥ 6 mm exposed ; 8 mm if main) ; slenderness limits as Unit II (180 compression, 350 reversing, 400 tension).

Effective length of compression members in trusses (IS 800 Cl. 7.5)

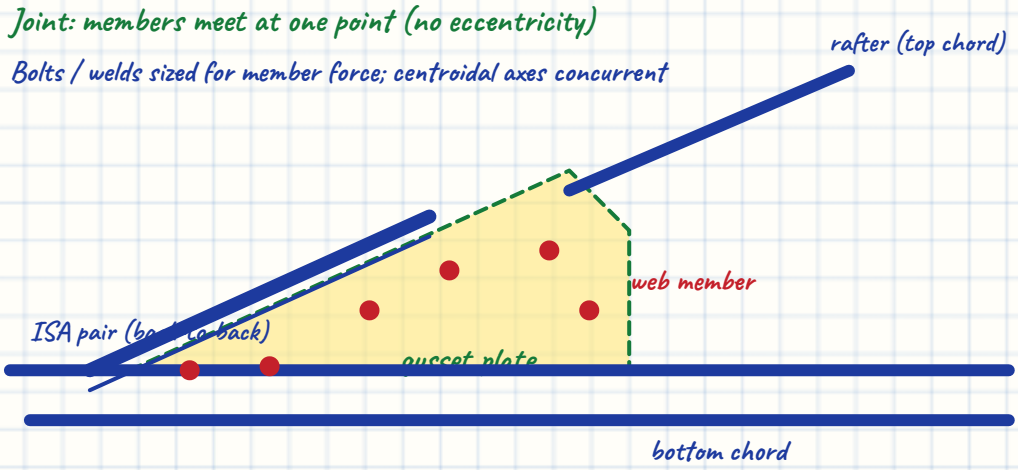
Member	Axis	Le/L (bolted / welded)
Chord – in-plane (v–v, between joints)	v–v	0.85 L
Chord – out-of-plane	y–y	1.0 L (distance between lateral supports, purlins/bracing)
Web (single angle)		0.7 L – 1.0 L

Two angles (back-to-back, tack-bolted) behave as a **T-section** — tack bolts at ≤ 600 mm. Gusset thickness 8–12 mm (by largest force).

Design of a compression chord (procedure)

- Assume $f_{cd} \approx 100$ MPa → $A_{req} = P_u/f_{cd}$; select 2 ISA ; (r_{xx} for in-plane, r_{yy} with gusset spacing for out-of-plane).
- $\lambda = \max(0.85\,L/r_{xx},\ L_y/r_{yy})$; **buckling class c** → f_{cd} → $P_d \geq P_u$. For double angles connected by tack bolts, check the slenderness of a single angle between tack bolts (≤ 0.7 × overall λ, tack bolt pitch ≤ 600 mm).
- **Tension member** (bottom chord): T_{dg}, T_{dn} (shear lag), T_{db} as per Unit II.

5. Design of Joints & End Bearings



Typical truss joint – members meeting at gusset plate

Joints

- Members joined with **gusset plate** (8–12 mm), bolts (M16–M20, 4.6/8.8) or fillet welds.
- **Centroidal axes of members meet at one point** (avoid eccentricity). For double angles, gusset passes between.
- **Number of bolts** for each member : n = Force/(bolt value) ; **bolt value** = min(shear (single/double), bearing on gusset or angle) ; or **weld length** = Force/(0.7 s f_{wd}) split between two angle legs (inversely with distances from centroid).

- Weld on angle legs: $L_1 = P\ y_2/(y_1+y_2)/(\text{strength})$, $L_2 = P\ y_1/(y_1+y_2)/(\text{strength})$

- Check **block shear** on gusset & angles; check gusset plate for tearing.
- Heel joint (rafter + bottom chord at support): gusset + bearing plate; **end bearing stiffener**.
- Ridge joint : both rafters connected via gusset or splice plate ; provide web members at purlin nodes.
- **Splice** of chords : cover plates for full force; staggered.

EXAMPLE 2 – WELDED JOINT OF ISA 75×75×8 TIE TO GUSSET

Force T_u = 150 kN, shop weld s = 6 mm (f_{wd} = 189.4) → strength = 0.7×6×189.4 = **795 N/mm**.

Required L_w = 150 000/795 = 189 mm. Distance of centroid from back: 21.4 mm → y_1 = 21.4, y_2 = 75 − 21.4 = 53.6.

L_1 (nearer heel) = 189 × 53.6/75 = **135 mm** ; L_2 = 189 × 21.4/75 = **54 mm** → plus 2 × 6 mm end returns. Provide L_1 = 140+, L_2 = 60 mm.

End bearings (supports of truss)

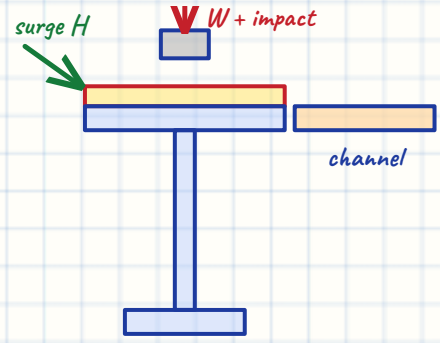
- **Fixed end** (one support) : base plate on masonry/concrete with **anchor bolts** (2 No.) ; area = R/(0.45 f_{ck}) (or masonry bearing 0.4–1 MPa); thickness for cantilever projection:

$$t = \sqrt{(2.5\ w\ (a^2 - 0.3b^2)\gamma_{m0}/f_y)} \text{ or } t = \sqrt{(3\ w\ a^2\gamma_{m0}/f_y)}$$

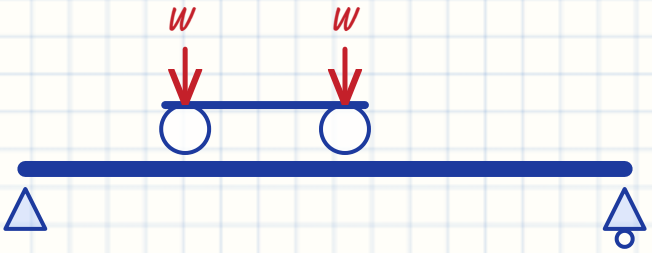
- **Roller / sliding end** (other support) for **L > 12–15 m** to allow thermal expansion — slotted holes in base plate or rollers/PTFE pads.
- Bearing on **bed block** (concrete) to spread reaction. Check **bearing stress**, anchor bolts for uplift (wind), and the bolt shear.

6. Gantry (Crane) Girder

Girder carrying an **overhead travelling (EOT) crane** wheel loads. Supported on **brackets of columns** ; span = column spacing (6–10 m typical).



Plan: rail + I-section + top channel
(channel resists lateral surge)



M_max under a wheel when span-centre bisects wheel & resultant

$M = W(2L - a)^2 / 8L$

Gantry girder section and moving wheel loads

Loads (IS 875 Part 2 / IS 800)

- **Vertical** : dead (rail ~ 0.3–0.5 kN/m, girder self-wt), **maximum wheel loads** from crane (crane + crab + lifted load) with **impact** : **25 % (EOT)**, 10 % (manual/pendant)
- **Lateral (surge)** : **10 %** of (weight of crab + lifted load) – shared by wheels on one side (EOT)
- **Longitudinal (braking)** : **5 %** of max. wheel load

Load combos: 1.5 [DL + (vertical + impact) + lateral] ; check each separately.

Section

- **Symmetrical I** (ISMB/ISWB) with **channel on top flange** (resists lateral), or plate girder for heavy cranes, or box.
- Rail fixed on top flange with clips; rail + flange act together.

Design steps

- 1. **Maximum B.M.** from moving loads (position by influence line) : for two equal wheels W at spacing a on span L:

$M_{max} = W(2L - a)^2 / (8L)$ (under one wheel when centre bisects wheel & resultant, $a < 0.586\,L$)

Max shear = reaction when wheel next to support : $V = W + W(L - a)/L$.

- 2. M_u (from vertical + impact + self-wt) and M_y (lateral, resisted by top flange + channel).
- 3. **Select section** : Z_{req}; **LTB check** with L_{LT} = span (compression flange laterally supported only at the ends, unless surge girder or bracing).
- 4. **Check combined bending**: $M_z/M_{dz} + M_y/M_{dy} \leq 1.0$.
- 5. **Shear** $V \leq V_d$ (low shear) ; **web buckling / crippling** under wheel (b_1 = wheel contact + rail dispersion at 45° through rail & flange).
- 6. **Deflection (vertical, w/o impact)** : L/750 (EOT ≤ 50 t), L/500 manual, L/1000 > 50 t ; lateral deflection L/ 1000-ish.
- 7. **Fatigue** (cyclic loads) – welded connections, bracket & **end connections to column** (bolted with bearing stiffeners).
- 8. Check **bracket** and **column** for eccentric load.

EXAMPLE 3 – MAX. B.M. FOR CRANE WHEELS

EOT crane : two wheels 100 kN each (including 25 % impact) at spacing a = 3 m on a span L = 6 m gantry girder. Check a < 0.586 L = 3.5 m ✓ (two-load rule applies).

$$M_{max} = 100 \times (2 \times 6 - 3)^2 / (8 \times 6) = 100 \times 81/48 = 168.75 \text{ kN·m}$$

Self-weight + rail 1.5 kN/m (service) → $M = 1.5 \times 6^2/8 = 6.75$ kN·m (≈ at centre).

Taking the 100 kN wheel loads as service loads (with impact) : $M_u \approx 1.5\ (168.75 + 6.75) =$ **263 kN·m**.

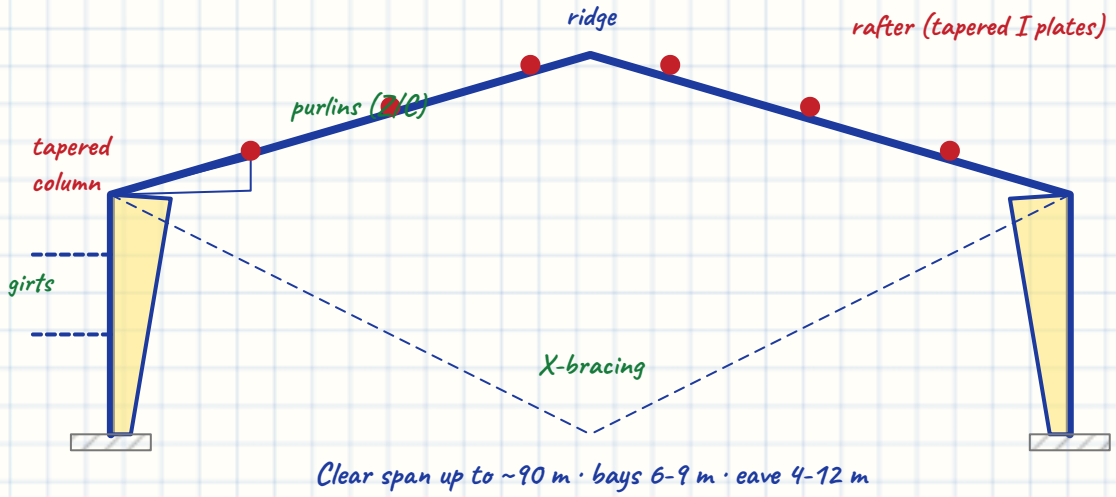
Lateral surge H = 10 % × (weight of crab + load = say 100 kN) = 10 kN per girder (shared by 2 wheels) → lateral moment $M_y \approx$ 10–15 kN·m, resisted by the top flange + channel.

Required $Z_p \geq 263 \times 10^6 \times 1.1/250 = 1157\ cm^3$ plus lateral → **ISWB 450 + ISMC 300 channel on top** (to be checked for biaxial bending & LTB).

> Maximum shear occurs with one wheel at the support; maximum deflection roughly at mid-span (wheel loads near centre).

7. Pre-Engineered Buildings (PEB)

PEB = steel buildings **designed, fabricated in the factory and bolted at site** from standard, optimised components – used for warehouses, factories, showrooms, hangars, metro stations.



PEB rigid frame with tapered members

Components

- **Primary framing** : rigid frames of **built-up tapered I-sections** (welded plates, web as thin as 5–6 mm, depth varies with bending moment) – columns, rafters, **haunch** at eave
- **Secondary framing** : **cold-formed Z / C purlins** (IS 801 / AISI), eave struts, **girts**, flange braces, sag rods
- **Roof & wall sheeting** : galvalume / colour-coated profiled sheets with insulation ; skylights, ventilators
- **Bracings** (X-bracing / rod / angle) for lateral loads ; **mezzanine floors, crane girders** can be added.
- **Foundations** : isolated footing with anchor bolts (pin or fixed base).

Advantages

- **Less steel** (≈ 30 % lighter than conventional) as depth follows moment diagram ; large clear spans (up to ~ 90 m) ; **quick erection** (weeks), pre-designed software (MBS, STAAD) ; economical, expandable ; good quality control.
- **Limitations**: limited to standard shapes; difficult for heavy crane loads (> 20–30 t) without customisation; fire protection, thin members → corrosion maintenance.

Design (outline)

- Loads : DL, LL (0.57 kN/m² typ. — IS 875) , **wind** (IS 875-3), seismic (IS 1893), crane, collateral. Combinations as IS 800 / MBMA.
- Analysis by **2-D frame software** ; design members by **IS 800 / AISC** for **web slenderness** and **tension-field action**; check deflection (**H/150 ; L/180–L/240**), local buckling.
- Connections : **end-plate moment connections with HSFG bolts** at haunch & ridge ; bolted base plate with anchor bolts.
- Codes: IS 800, IS 875, IS 801 (cold-formed), MBMA (Metal Building Manufacturers Association) manual.

Conventional vs PEB

Item	Conventional	PEB
Members	hot-rolled sections	tapered welded plates + CFS purlins
Weight	higher	25–30 % lower
Design/fabrication time	long	short, standardised
Site work	high	bolting only
Flexibility	high	moderate

8. Worked Examples – Truss

Example 4 – Wind load on a roof (IS 875 Part 3 style)

Basic wind speed V_b = 44 m/s ; $k_1 = k_2 = k_3 = k_4 = 1.0$ (assumed) → V_z = **44 m/s**

$p_z = 0.6 \times 44^2 = 1162$ N/m² = **1.16 kN/m²** ; $p_d = K_d\, p_z = 0.9 \times 1.16$ = **1.045 kN/m²** (K_a K_c taken as 1.0, conservative)

Roof slope 20° (low-rise): leeward / windward suction $C_{pe} \approx -0.7$ (read exact value from IS 875 Table) ; medium openings $C_{pi} = \pm 0.5$.

Worst net : $C_{pe} - C_{pi} = -0.7 - (+0.5) =$ **−1.2** → $F = -1.2 \times 1.045 =$ **−1.25 kN/m²** (suction, ⊥ roof).

Truss spacing 4 m, purlin spacing 1.5 m → joint (purlin) uplift = $1.25 \times 4 \times 1.5$ = **7.5 kN** per purlin ; factored $1.5 \times 7.5 = 11.3$ kN. Check 0.9 DL + 1.5 WL : **net uplift** → members in bottom chord & webs may reverse (tension ↔ compression) ⇒ design for both.

Example 5 – Member forces + design (symmetrical roof truss)

Span 12 m, rise 3 m → $\tan\theta = 0.5$, $\theta = 26.57°$, rafter length = $\sqrt{(6^2 + 3^2)} = 6.71$ m, 2 panels per rafter (3.35 m). Factored load = 1.5 kN/m² (plan) × 4 m spacing × 12 m = 72 kN → joint loads (plan panels 3 m): eave 9, mid 18, ridge 18, mid 18, eave 9 kN. Reactions R = 36 kN each.

Joint at left support (only rafter + bottom chord meet) : net vertical = 36 − 9 = 27 kN

> Rafter (end panel) : $C = 27 / \sin\theta = 27/0.447 =$ **60.4 kN (compression)**
> Bottom chord (end panel) : $T = C\cos\theta = 60.4 \times 0.894 =$ **54.0 kN (tension)**

Check by sections at mid-span : M about ridge = 36×6 − 9×6 − 18×3 = 108 kN·m → central bottom chord = 108/3 = 36 kN (T) ✓ (smaller than the end panel – consistent with a Fink/Howe web pattern).

Rafter design : P_u = 60.4 kN, L = 3.35 m, 2 ISA 65×65×6 back-to-back (A = 1488 mm², $r_{xx} \approx 19.7$ mm from tables)

$\lambda = 0.85 \times 3354 / 19.7 =$ **145** < 180 ✓ ; class c ($\alpha = 0.49$) : $f_{cc} = \pi^2E/145^2 = 93.9$; $\bar{\lambda} = 1.632$; $\varphi = 2.18$; $\chi = 0.275$

$f_{cd} = 0.275 \times 250/1.1 = 62.6$ MPa → $P_d = 62.6 \times 1488 =$ **93 kN > 60.4** ✓

Bottom chord T_u = 54 kN : $A_g(req)$ = 54×10³×1.1/250 = 238 mm² ; provide minimum **2 ISA 50×50×6** → check T_{dn}, T_{db} and slenderness (350) ✓ (governed by minimum size).

EXAMPLE 6 – BEARING PLATE FOR THE FIXED END

R_u = 36 kN on M20 concrete pedestal : $f_{bearing}$ = 0.45 × 20 = 9 MPa → A_{req} = 36 000/9 = 4000 mm² → provide practical **200 × 200** plate (area 40 000, w = 0.9 N/mm²) ; projection a = 75 mm

$t = \sqrt{(3\ w\ a^2\ \gamma_{m0}/f_y)} = \sqrt{(3 \times 0.9 \times 75^2 \times 1.1/250)}$ = **8.2 mm → 10 mm** plate ; 2 anchor bolts M16 ; roller end : slotted holes 20 × 40.

9. Bracing, Girts and Stability of the Shed

Bracings in an industrial shed

- **Roof (horizontal) bracing** – X-bracing in the plane of rafters/bottom chord at end bays & every 4th–5th bay : transfers wind from gable columns to side walls and gives lateral support to top chord (compression) → reduces L_y.
- **Vertical bracing between trusses** (at ridge, eaves) – prevents overturning/lateral buckling of trusses during erection.
- **Longitudinal bracing in columns** – resists wind/crane braking along the length.
- **Knee braces / portal action** for lateral stiffness of frame.
- Bracing members : angles, flats, rods (tension only – designed for tension, 2 rods per panel).

Girts (sheeting rails)

Horizontal members on wall columns carrying cladding & wind ⊥ to wall: designed like purlins (bending about both axes; sag rods to resist vertical self-weight/sheeting). Deflection limit L/150 (elastic) or L/180.

Column of industrial building (just idea)

- **Stepped column** with crane bracket : lower part carries crane + roof; upper part carries roof only.
- Design for axial + bending (IS 800 Cl. 9.3) ; base connected with anchor bolts (fixed/pinned).
- Wind on walls → eave reactions ⇒ horizontal load on column; frames analysed for portal action.

Angle purlin – quick checks (small spans)

- Use **ISA 100×75×8 / 90×90×8** for spans ≤ 4 m, long leg ⊥ to roof ; connect to rafter by **cleat/ angle** (bolted).
- Moment M_x about the geometric axis parallel to roof, M_y about the other : angle is **un-symmetric** → bending about principal axes (u-u, v-v); principal moment components : $M_u = M_x \cos\alpha + M_y \sin\alpha$ ...
- Use gross elastic section modulus with 1.2 limit; check deflection (≈ governing).

Channel purlin – quick checks

- ISMC 100–150 with flanges toe-up/ toe-down; shear centre issue → torsion small if sheeting connected to top flange.
- Good for spans 4–6 m; **web crippling** at the bearing on the rafter, **local bearing** with cleat.
- Cold-formed **Z / C purlins** (IS 801) with lap splices are used for spans 6–9 m (PEB).

Important formulae – Unit IV (quick sheet)

$$w_n = w\cos\theta\ ;\ w_p = w\sin\theta\ ;\ M_x = w_n L^2/10 \text{ (cont.)} / w_n L^2/8 \text{ (SS)}$$

$$p_z = 0.6\, V_z^2\ ;\ F = (C_{pe} - C_{pi})\, A\, p_d$$

$$\text{LL on roof} = 0.75 - 0.02(\theta - 10)\ \text{kN/m}^2 \geq 0.4$$

$$\text{Joint load} = (\text{kN/m}^2) \times \text{truss spacing} \times \text{purlin spacing (plan)}$$

$$M_{max} = W(2L - a)^2/(8L) \text{ (2 equal wheel loads)}\ ;\ \text{vertical impact 25 \% (EOT)}$$

$$\text{Surge} = 10\ \%\ (\text{crab} + \text{load})\ ;\ \text{Longitudinal braking} = 5\ \%\ \text{wheel load}$$

$$t_{base} = \sqrt{(3\, w\, a^2 \gamma_{m0} / f_y)}\ \text{(cantilever base plate)}$$

Important exam questions

1. Explain different types of roof trusses, with neat sketches.
2. Calculate wind load on a pitched roof truss (IS 875) and panel-point loads.
3. Design an angle purlin for a roof with given loads / design a channel purlin with sag rods.
4. Design a Fink truss member (rafter & tie) and the welded joint at the heel.
5. Explain loads on gantry girders and design a gantry girder for a 10 t EOT crane.
6. Write short note on PEB – components, advantages, design considerations.